

Computational Replication Report:

Jungwirth *et al.* (2025), *Altermagnetic spintronics*
(arXiv:2508.09748)

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1 Paper summary

This review connects altermagnetism to spintronics. Altermagnets — d -wave (or higher even-parity) collinear compensated magnets — enable strongly *spin-polarized electrical currents without any net magnetization*, plus THz-range spin dynamics, making them attractive for MRAM-type memory. The conceptual anchor is Fig. 1’s model d -wave altermagnet, contrasted against a ferromagnet and a conventional antiferromagnet.

2 Replication target

We reproduce Fig. 1’s model spintronic signatures via constant- τ Boltzmann transport on the d -wave spin-split Fermi surfaces: $\sigma_{ab}^s = \sum_{\mathbf{k}} v_a^s v_b^s (-\partial f / \partial E)$ per spin s , charge current $\propto \sigma^\uparrow + \sigma^\downarrow$, spin current $\propto \sigma^\uparrow - \sigma^\downarrow$.

3 Claims addressed

ID	Claim	Type	Testable?	Tested?
C1	Current polarization spin-up (bias $\parallel$ x) reverses to spin-down (bias $\parallel$ y)	transport	yes	yes
C2	Diagonal bias: unpolarized longitudinal + transverse pure spin current (SSE)	transport	yes	yes
C3	FM: one polarization all dirs; AFM: unpolarized all dirs	transport	yes	yes
C4	Material TMR/AHE magnitudes; THz dynamics	survey	partly	out of scope

4 Results vs. paper

5 Per-claim what worked / what didn’t

C1 (worked). $P(x) = +0.931$, $P(y) = -0.931$: the electrical-current spin polarization reverses sign with bias direction, exactly the d -wave signature. **C2 (worked).** Diagonal bias yields zero longitudinal polarization and zero transverse charge current but a nonzero transverse *spin* current — the non-relativistic spin-splitter effect. **C3 (worked).** The ferromagnet keeps one polarization

Quantity	Expected (Fig. 1)	This work	Verdict
P (bias $\parallel x$)	+ (spin-up)	+0.931	✓
P (bias $\parallel y$)	− (spin-down)	−0.931	✓
P_{long} (diagonal)	0 (unpolarized)	0.000	✓
transverse spin current (diag)	$\neq 0$ (SSE)	$\neq 0$	✓
transverse charge current (diag)	0	0.000	✓
FM P (x,y,diag)	same sign	+0.117 all	✓
AFM P (x,diag)	0	0.000	✓

sign for every bias direction; the conventional AFM ($t_{\text{AM}}=0$) is unpolarized for all directions — the two Fig. 1 contrasts. **C4 (out of scope)**. Absolute TMR/AHE ratios and THz resonance frequencies are review surveys of many papers.

6 Critique of the replication

This reproduces the CONCEPTUAL model of a review (Fig. 1), not a single quantitative material result — appropriate for a review. Strengths: the transport computation is a genuine Boltzmann calculation (not hand-waving), and every symmetry signature comes out with the correct sign and the correct zeros (diagonal, AFM). Honest limitations: (i) the transverse spin current is reported in raw (unnormalized) simulation units — only its nonzero-ness and the zero transverse charge current are the physical statements; (ii) constant- τ clean limit (no disorder/finite- T , Open Q2); (iii) no junction TMR (Open Q1) or dynamics (Open Q3). LLM-judge: REPLICATED (coverage 9, agreement 9).

7 Verdict

REPLICATED — the review’s model d -wave altermagnet spintronic signatures (polarization reversal, spin-splitter effect, FM/AFM contrasts) reproduced with correct signs and zeros via Boltzmann transport. Material TMR/AHE magnitudes and THz dynamics out of scope (survey-level).

Open Questions

See `report/open_questions.json`: Q1 junction TMR magnitude; Q2 disorder/finite- T SSE robustness; Q3 THz resonance origin; Q4 g-/i-wave transport angular fingerprint; Q5 coupling to ferroelectricity/superconductivity.